

Seminar Presentation on

Wind Profile Estimation for K-Band Doppler Radar Based on Mutual Convolution Cost Function Method

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1. Background & Introduction

1.1 Background

- Wind profiles are estimated from Doppler Radars via signal processing techniques.
- Radar echoes are often heavily corrupted by background noise (gets worse in cold climate)
- Traditional methods (like coherent accumulation) works well only when SNR is high
- Under low SNR conditions, multiple peaks are observed

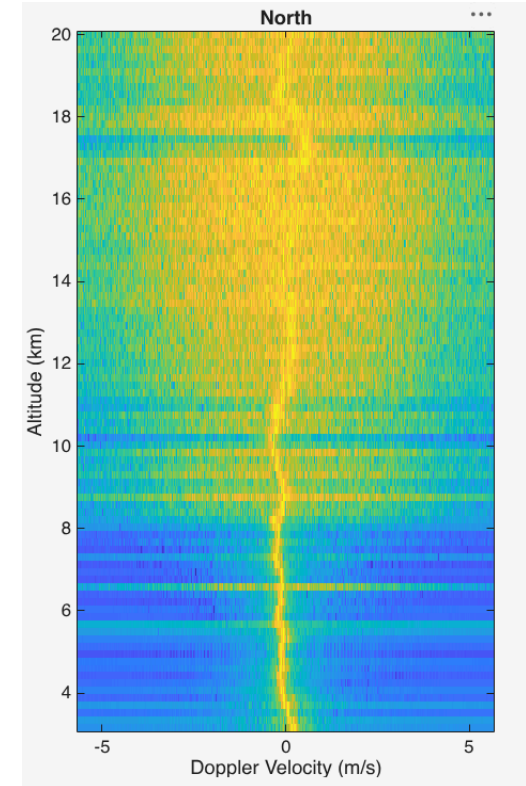


Figure 1: Doppler spectrum of an observation

1.2 Introduction

- Paper proposes a new method to estimate wind profiles from doppler radars by combining several methods including MCCF.
- Works well under low SNR and multiple peaks conditions.
- The paper primarily focuses K-Band radar and near surface wind profile and on its potential application to efficient wind energy generation.

2. Experimental Setup

2. Experimental Setup

- Experiment equipment: K-band vortex radar (VORTRAD)
- Ground truth: LIDAR (usually has 99% accuracy in clear sky)
- At the end, correlation among LiDAR and Radar is obtained



Figure 2: Experiment Setup

2.1 Overview

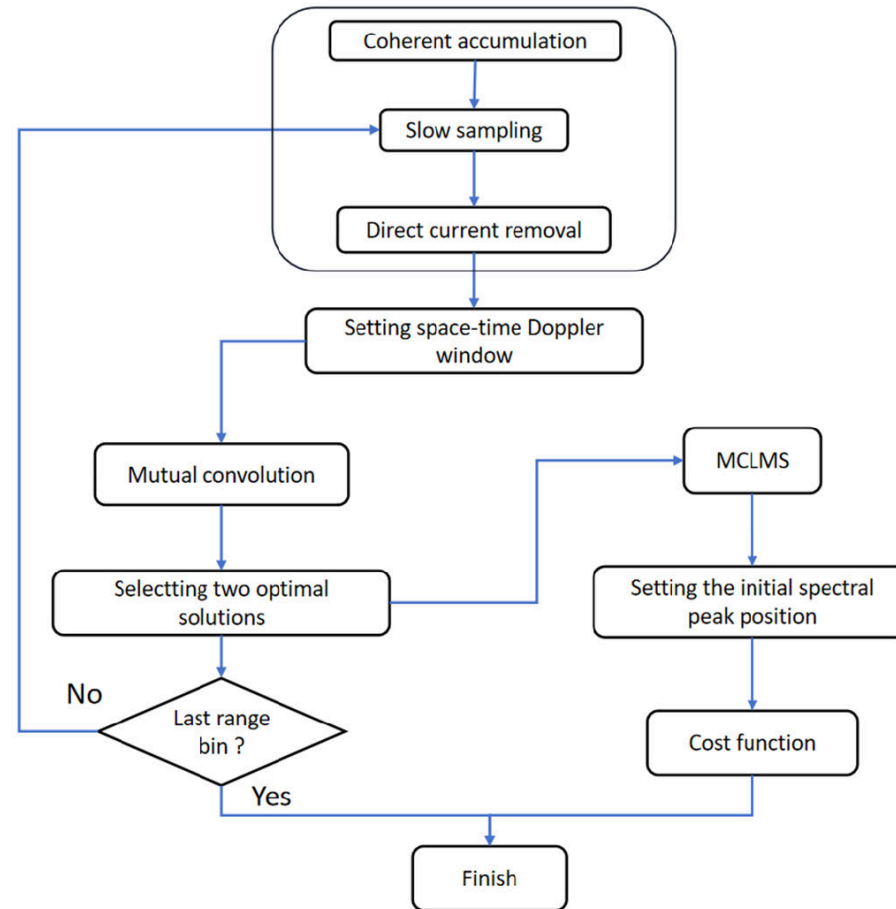


Figure 3: Flow diagram of method

3. Proposed Method

3.1 Signal Preprocessing

3.1.1 Coherent Accumulation

The CA adds and averages short segments of radar signals improving SNR
Result after N-order CA is represented as:

$$X = [x_{c1}^T, x_{c2}^T, \dots, x_{c(N-1)}^T, x_{cN}^T]$$

Where x_{ci} is result of i^{th} CA.

3.1.2 Slow Sampling

- On every Pulse Repetition Interval (PRI), a pulse is transmitted, radar receives echoes from all range bins (heights)
- For a single height, there will be 1 echo sample from every transmitted pulse.
- To obtain this, a row vector from X is used. This process is called *slow sampling* in the distance dimension

3.1 Signal Preprocessing

3.1.3 DC and Clutter Removal

- VORTRAD produces large DC power and ground clutter due to its zero-IF RF transceiver architecture.
- Here these are removed.
- The paper doesn't mention a particular clutter removal method

3.2 Setting Space-Time Doppler Window

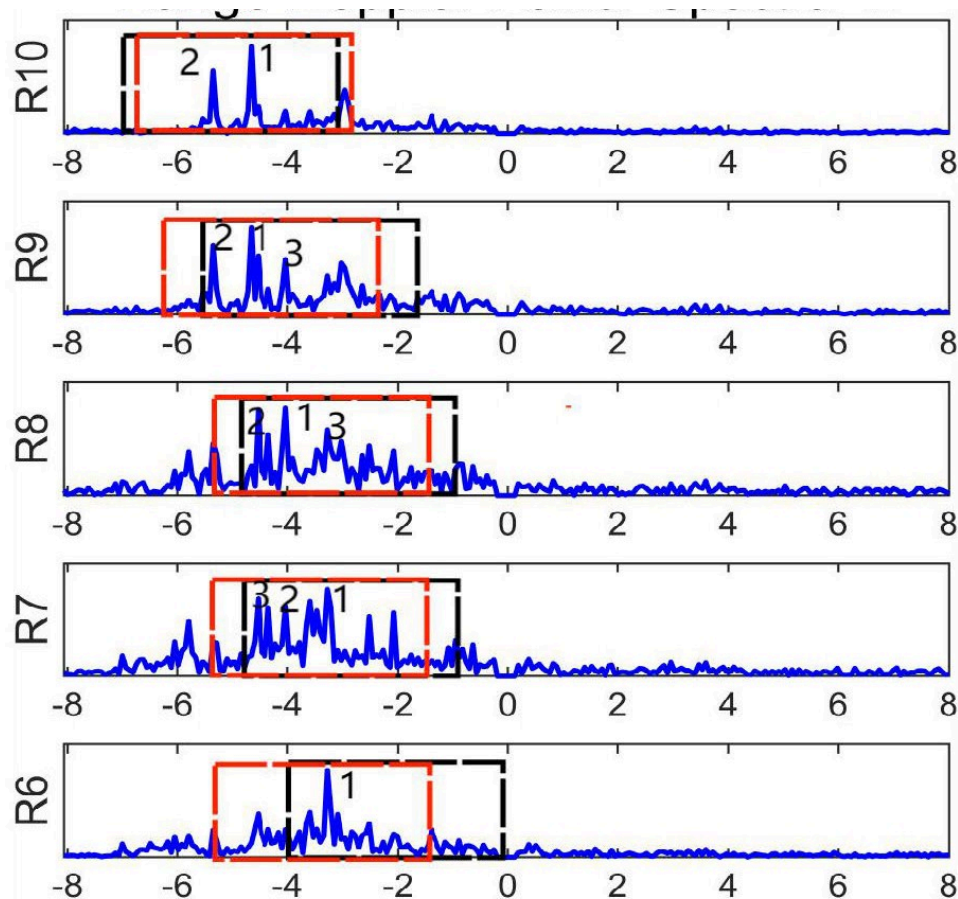


Figure 4: ST-W Window

- Idea is to restrict search region in both space and time (ST-W) for finding candidate peaks
- Assumes wind velocity changes smoothly with altitude (space) and between successive radar scans (time).
- In the figure
 - ▶ Red -> Space window
 - ▶ Black -> Time window

3.2 Setting Space-Time Doppler Window

Space window: $W_1 = (F_0 - \alpha d_f, F_0 + \alpha d_f)$

Time window: $W_2 = (T_0 - \alpha d_f, T_0 + \alpha d_f)$

$$\text{Space Time window: ST-W} = \begin{cases} W_1 \cap W_2 & \text{if } W_1 \cap W_2 \geq \alpha d_f \\ W_1 & \text{else} \end{cases}$$

Where

$$\text{wind shear } d_f = \frac{V_d}{d_v} = \frac{V_d}{f_{d\min} \lambda / L}$$

V_d = maximum value of wind shear

L = number of points in FFT.

α = window coefficient

F_0 and T_0 : Space and time spectrum estimation
result of the previous range bin

3.3 Mutual Convolution

- Main idea is to use the temporal correlation of echoes to suppress uncorrelated noise.

Method:

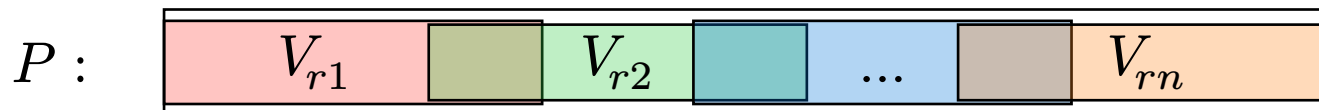
1. Take a row vector $P = [p_1, p_2, \dots, p_N]$ of one range bin from X
2. Create new vectors from P with equal length (say half size)

$$V_{r1} = [p_1, p_2, \dots, p_{N/2}]$$

3. From this a new vector V can be written as:

$$V = [V_{r1}, V_{r2}, \dots, V_{rn}]$$

Where V_{ri} is segment of datas from different intervals in same range bin



3.3 Mutual Convolution

4. Mutual convolution is performed:

$$\begin{aligned}V_{C1} &= V_{r2} * V_{r1}^{\text{HR}} \\V_{C2} &= V_{r3} * V_{r1}^{\text{HR}} \\&\vdots \\V_{C(n-1)} &= V_{rn} * V_{r1}^{\text{HR}}\end{aligned}$$

x^{HR} → take conjugate and then reverse order

5. Extracting the middle of the vector (To “maintain the same time complexity and velocity resolution of calculation”):

3.3 Mutual Convolution

$$[V'_C] = \begin{bmatrix} V'_{C1} \\ V'_{C2} \\ \vdots \\ V'_{C(n-1)} \end{bmatrix} = \begin{bmatrix} V_{C1} \left(\frac{N}{4} : \frac{3N}{4} \right) \\ V_{C2} \left(\frac{N}{4} : \frac{3N}{4} \right) \\ \vdots \\ V_{C(n-1)} \left(\frac{N}{4} : \frac{3N}{4} \right) \end{bmatrix}$$

6. Then maximum of power spectra of each spectrum is computed:

$$V'_{cf} = \left[\max \left\{ \left| \mathcal{F}([V'_c]) \right| \right\} \right]$$

7. From V'_{cf} we take a optimal solution X_r corresponding to the “maximum power spectrum”. Also convolution results of adjacent delays are recorded as suboptimal solutions X'_r

3.4 MCLMS

- ~ An adaptive filter, pretrained.
- X_r and X'_r are taken input and desired signal of the filter.
- After the coefficients have been fully updated, filter excels at the recognizing stable signals and removes multiple strong clutter.

For finding Filter coefficients:

1. Calculating the normalized parameters of filter coefficients as:

$$u = 1 / \left(\sum_{j=1}^{N/2} |X_r(j)|^2 \right)$$

3.4 MCLMS

2. X is constructed from the samples of X_r :

$$X = \begin{cases} \text{zeros}(1, k - j, X_r(1 : j)) & , j < k \\ X_r(j - k + 1 : j) & , j \geq k \end{cases} \quad j \in \{1, 2, \dots, N/2\}$$

3. Initializing filter coefficients as $W_1 = 0$. The coefficients are then updated using:

$$Y_j = W_1 X^T$$

$$e_j = X'_{rj} - Y_j$$

$$W'_1 = 0.98W_1 + ue_j X^H$$

4. Repeating above steps until all data are filtered, the coefficients have been fully updated

5. We will use this coefficient to obtain the filtering result Y_r

3.5 Generating the Cost Function

- To find out *candidate peaks*.

Steps

1. Calculate the average value corresponding to the power spectrum of the first range bin and pick out the prominent peak:

$$M_{1,1} = \max \left| \frac{1}{n-1} \sum_{i=1}^{n-1} \mathcal{F}(V'_{Ci}) \right|$$

Use this as starting point for ST–W

2. Compute doppler spectrum of each height bin

$$M_r = |\mathcal{F}(Y_r)|$$

3. In each ST–W, spectral peaks are sorted and up to 3 peaks are selected.

3.5 Generating the Cost Function

4. If $M_{r\max}$ is spectral peak, then candidate peaks are:

$$M_{r,j} \geq 0.4M_{r\max}$$

5. Cost value for each peak is:

$$V_{r+1,j} = 1 - \left| \frac{(M_{r+1} - M_r)d_v}{\alpha V_d} \right|$$

d_v = velocity resolution

6. MCCF cost function:

$$F_{r+1,j} = w_1 V_{r+1,j} + w_2 M_{r+1,j} \quad \text{and } w_1 + w_2 = 1$$

w_1, w_2 are constraint weights

3.5 Generating the Cost Function

7. The peak with the highest cost value is selected
8. Repeat this for all height bins

4. Results

4.1 Performance at Different Delay Times

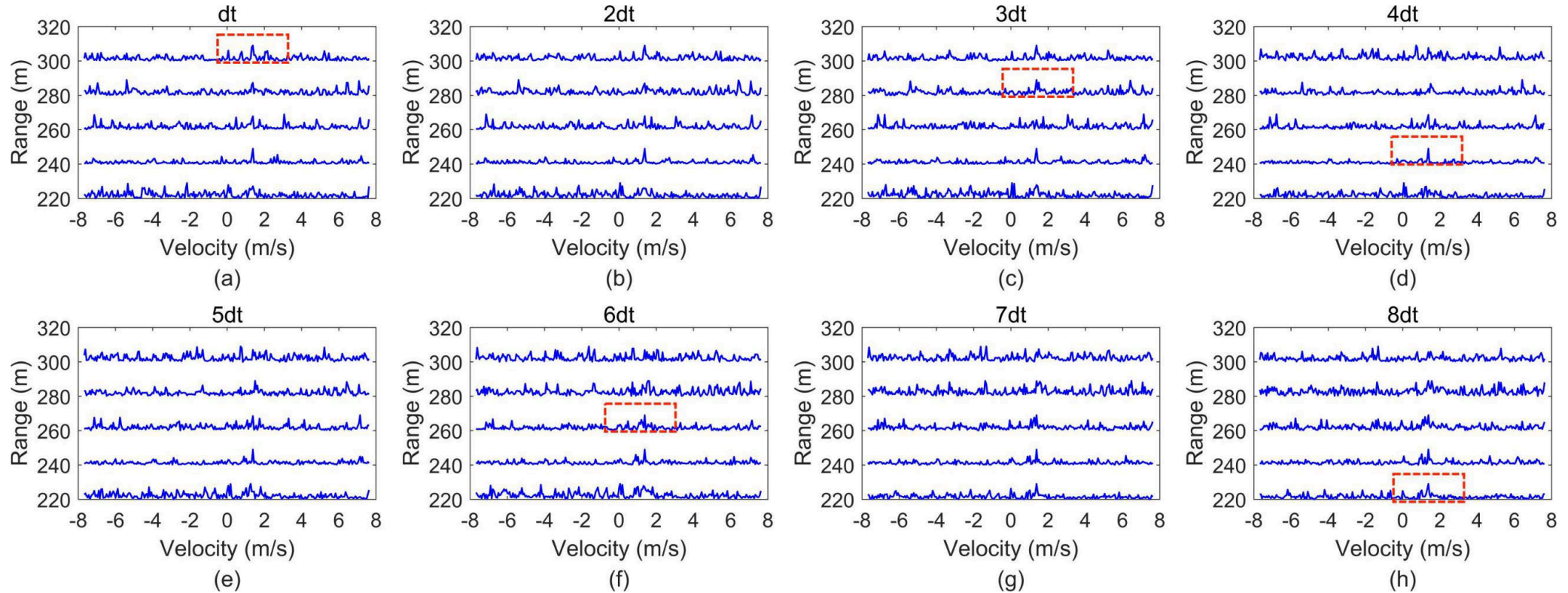


Fig: Optimal solutions are marked by red dotted box of different range bin after mutual convolution (a)–(h) Represent the filtering effect of dt – $8dt$ delay time respectively.

4.1 Performance at Different Delay Times

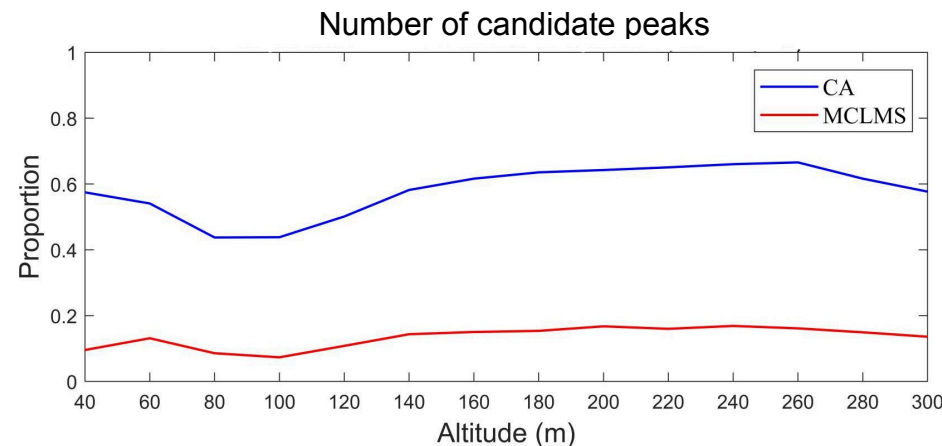
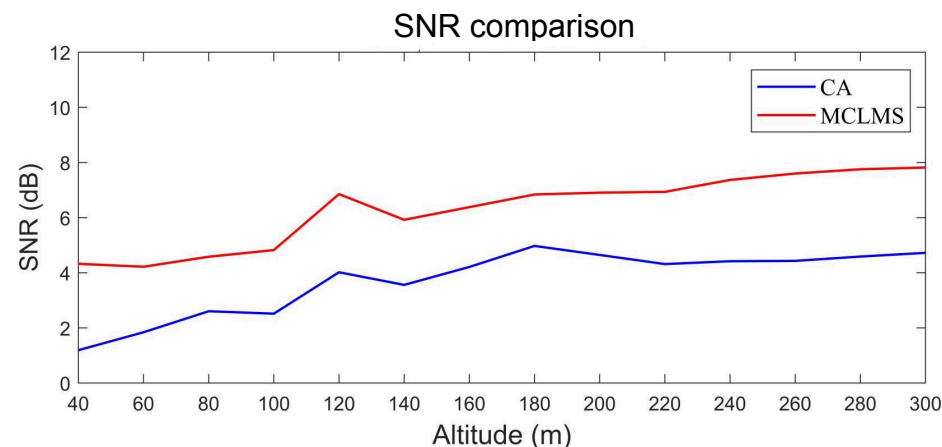
- As the delay time increases, the number of statistics gradually decreases
- Still not possible to determine a specific optimal delay time.

Delay	RI	R2	R3	R4	R5	R6	R7	R8
d_t	1543	1503	1568	1471	1486	1488	1498	1568
$2d_t$	444	404	430	420	398	397	406	403
$3d_t$	179	203	195	224	214	218	223	196
$4d_t$	128	129	91	119	114	127	122	126
$5d_t$	70	72	68	73	85	80	78	70
$6d_t$	39	63	58	69	75	71	66	35
$7d_t$	42	60	38	63	59	51	48	47
$8d_t$	55	66	52	61	69	68	59	55

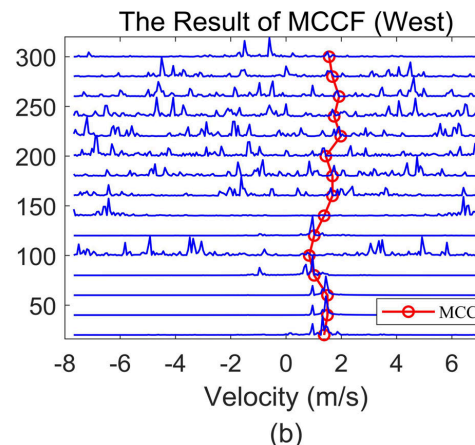
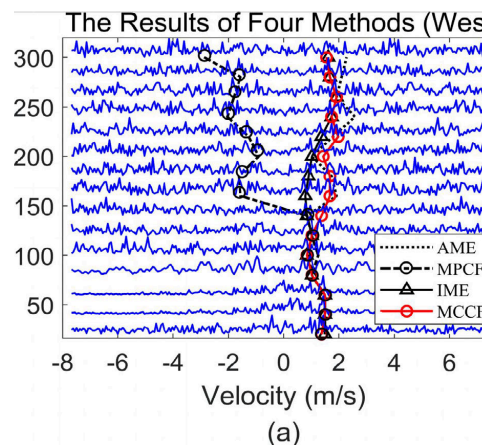
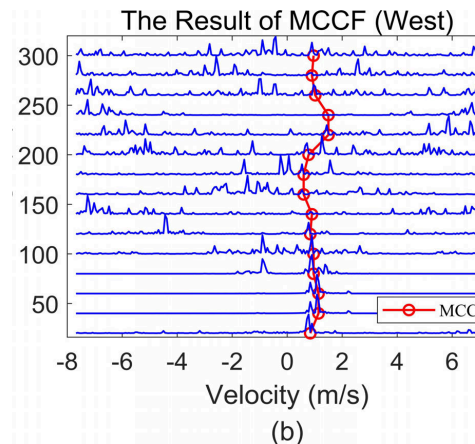
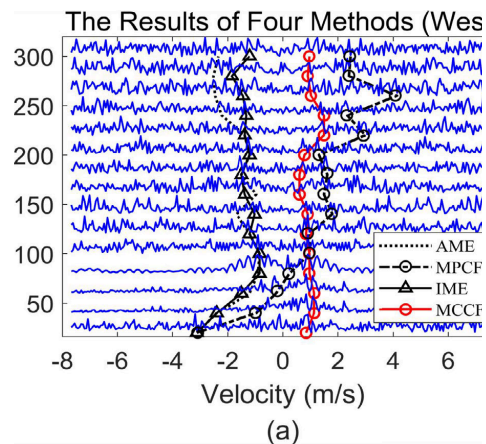
Table 1: Statistics of optimal delay time for mutual convolution

4.2 Comparison of Peak Number and SNR

- Average SNR for all bins plotted for CA and proposed MCLMS filters at different altitudes
- Proportion of spectra containing more than six candidate peaks



4.3 Wind Profile Estimation Results



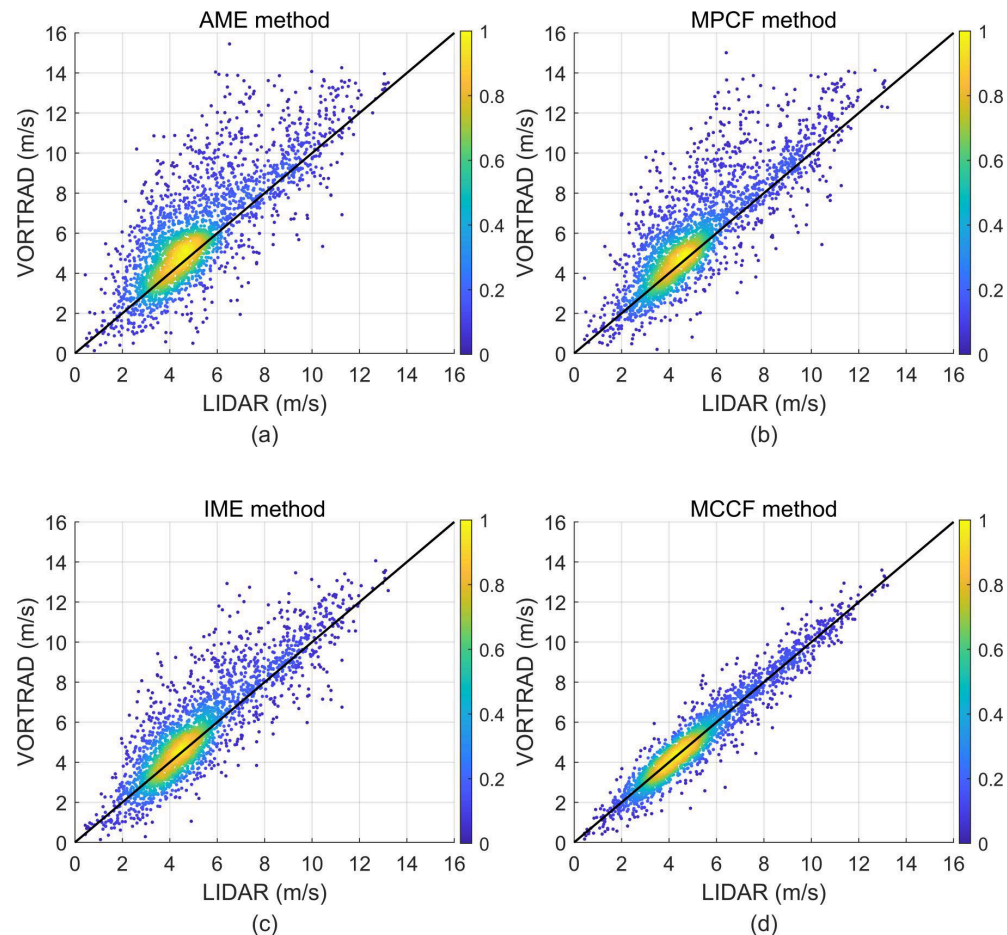
Comparison of four methods based on the data recorded on

- Dec 18, 2022 (15:19:32) (top row)
- Dec 18, 2022 (15:19:33) (bottom row)

4.4 Correlation with LiDAR observation

- IME shows some improvements over the AME and MPCF
- However MCCF aligns noticeably well with the LIDAR
- Correlation coefficients are summarized below:

Height(m)	AME	MPCF	IME	MCCF
40	0.8417	0.8368	0.8369	0.9576
80	0.8319	0.8350	0.8461	0.9609
120	0.8320	0.8431	0.8695	0.9625
160	0.8137	0.8353	0.8761	0.9641
200	0.7908	0.8276	0.8764	0.9647
240	0.7830	0.8202	0.8738	0.9650
280	0.7837	0.8219	0.8744	0.9651



5. Conclusion & Limitations

5. Conclusion & Limitations

- The proposed algorithm was applied on data collected by the VORTRAD K-band wind profiler and verified against LiDAR observation
- Method reduced false spectral peaks and improved the SNR ratio by approximately 2–4 dB
- Algorithm produces smoother and more continuous wind profiles
- Achieves highest correlation coefficient (0.96) with LiDAR observation.
- Authors note that the current method is validated only for clear-weather conditions and does not account for precipitation echoes such as rain or snow

Thank You